

- (a) If $r < \underline{r}$, an increase in homophily increases the virality of content.
- (b) If $r > \bar{r}$, a decrease in homophily increases the virality of content.

Theorem 2 shows how low-reliability content can spread virally in networks with high homophily. Intuitively, when content comes from a low-reliability source, only agents who (strongly) agree with the article’s message share it. However, as homophily increases, users know that they will mostly share with other like-minded people, who will also be inclined to share this content. This creates an echo chamber-like environment: the likelihood of being called out for spreading misinformation is now lower, making users “less disciplined” or more likely to share lower-reliability content. Strategic complementarities then extend these incentives throughout the network. In this way, homophily leads to the viral spread of low-reliability articles that likely contain misinformation.

However, Theorem 2 shows that homophily can have non-monotone effects. This is because greater homophily also keeps an article circulating among the same group of like-minded users and reduces the likelihood that it will reach other communities. Theorem 2(a) establishes that the first effect of homophily, working through incentives to share low-reliability content, is more powerful than the second, “circulation effect”, when we focus on particularly low-reliable content (with $r < \underline{r}$). This implies, in particular, that homophily’s impact is to increase the virality of especially low-quality content, which is of course relevant for public policy (as we discuss in Section 6).

The results in Theorem 2 are in line with recent evidence highlighting the importance of echo chambers for the spread of misinformation. Törnberg (2018) and Vicario et al. (2016), among others, show that homophily in sharing behavior propagates ideologically-congruent ideas, with little incentive to question the veracity of this information, while Quattrociocchi et al. (2016) document how echo chambers on Facebook fuel conspiracy theories and the popularization of incorrect scientific ideas, for example, on vaccines. Levy (2021) provides evidence that “filter bubbles” generated by Facebook’s algorithms are an important source of propagation of misinformation.

4.2 Comparative Statics: Divisive Content and Belief Polarization

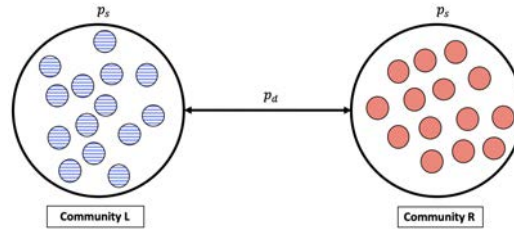


Figure 2. Two-Island Model.

For simplicity, for the next result, we focus on the case of just two islands, a left-wing and a right-wing one with prior distributions H_L and H_R , respectively, as pictured in Figure 2. Moreover, we suppose that there is disjoint support of prior beliefs across communities. Formally, we assume H_R has support on $[\underline{b}_R, \bar{b}_R]$ and H_L has support on $[\underline{b}_L, \bar{b}_L]$, with $\bar{b}_L < 1/2 < \underline{b}_R$.